

K-NN

K-Nearest Neighbors: Decision Boundaries

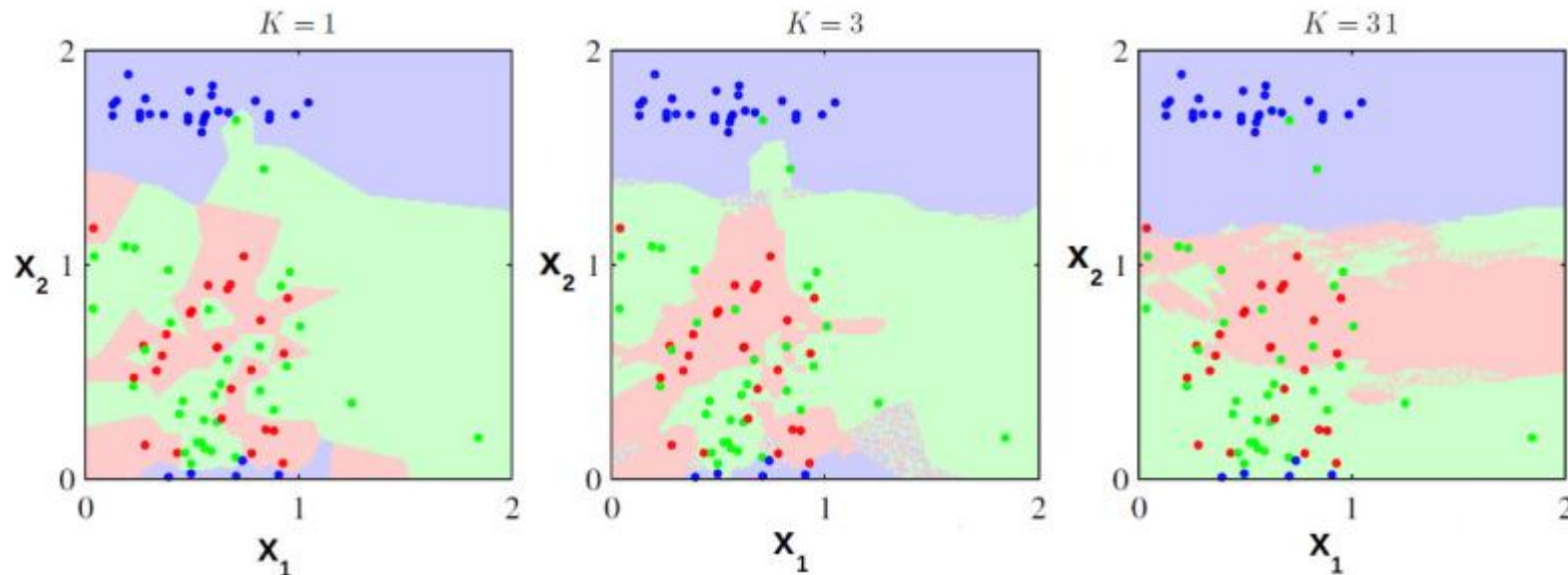
- A good value for K can be determined by considering a range of K values

- K too small: we will model the noise

- K too large: neighbors include too many points from other classes

Very $K \rightarrow$ obscure validation error
 K for which least validation error

- Larger K leads to smoother decision boundaries

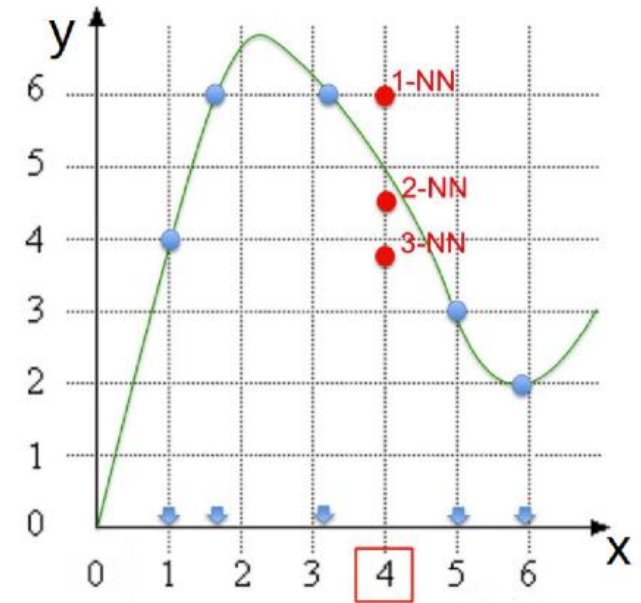


K-NN for Regression

- Given: training examples $\{(\mathbf{x}^{(1)}, y^{(1)}), (\mathbf{x}^{(2)}, y^{(2)}), \dots, (\mathbf{x}^{(n)}, y^{(n)})\}$
 - $\mathbf{x}^{(i)}$ attribute-value representation of examples
 - $y^{(i)}$ real-valued target (profit, rating on YouTube, etc.)
- Testing point $\mathbf{x}$ that we want to predict the target

Algorithm:

- compute distance $D(\mathbf{x}, \mathbf{x}^{(i)})$ to every training example
- select k closest instances $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(k)}$ and their labels $y^{(1)}, y^{(2)}, \dots, y^{(k)}$
- output the mean of $y^{(1)}, y^{(2)}, \dots, y^{(k)}$: $\bar{y} = \frac{1}{k} \sum_{j=1}^k y^{(j)}$



Distance fm. -

① Euclidean Distance:- x, x'

$$D(x, x') = \sqrt{\sum_d |x_d - x'_d|^2}$$

- Symmetric, spherical, treats all dimensions equally
- Sensitive to outliers.

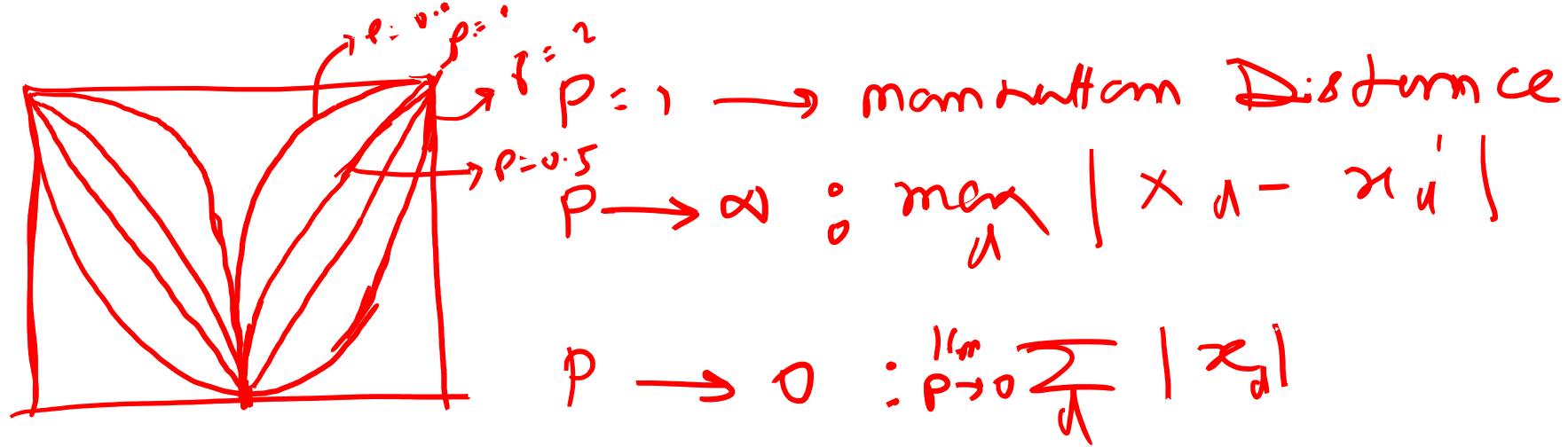
② Hamming Distance:- $D(x, x') = \sum_d \mathbb{1}_{x_d \neq x'_d}$

↳ Number of attributes where $x_d \neq x'_d$ differs

③ Minkowski Distance:- (p-norm)

$$D(x, x') = \sqrt[p]{\sum_d |x_d - x'_d|^p}$$

$p=2 \rightarrow$ Euclidean Distance



$p \leq 1 \rightarrow$ Promotes Sparsity.

(iv) Kullback - Leibler Divergence: -

For histograms ($x_d > 0, \sum_d x_d = 1$)

$$D(x, x') = - \sum_d x_d \log \frac{x_d}{x'_d}$$

Resolving Ties:-

- Equal no. of +ve / -ve neighbors.
- Breaking Rule:
 - Coin toss
 - Prior: $P(c_k) = \frac{N_k}{N}$
Choose larger $P(c_k)$
 - 1-NN

Data: $\{x_1, \dots, x_N\}$
 $\nwarrow$ $p(x)$

$\underbrace{R}_{k \text{ data points}}$ $P = \frac{k}{N}$

If R is very small $\rightarrow p(x)$ does not change much

$$\int_a^b p(x) dx \quad P \approx p(x) V \quad [\text{Volume of the region}]$$

$$\frac{k}{N} = p(x) V$$

$$p(x) = \frac{k}{N V}$$

$N_K \rightarrow C_K$ class

$$P(x | C_K) = \frac{k_K}{N_K V}$$

$$P(C_K) = \frac{N_K}{N}$$

$$P(C_K | x) = \frac{P(x | C_K) P(C_K)}{P(x)}$$

$$= \frac{\frac{k_K}{N_K V} \cdot \frac{N_K}{N}}{k/NV} = \frac{k_K}{k}$$